Stone Skipping Physics Simulator v2

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This document outlines the physics model behind my new and improved WebGL stone-skipping demo. It is a reference for the math, not a derivation. For the latter, please see Bocquet [Bocquet 2003] and the follow-up work [Clanet et al. 2004, Nagahiro and Hayakawa 2004, Bocquet et al. 2005]. The model differs from Bocquet in one notable way: the original model takes a fixed order-of-magnitude lever arm $OP \sim R$ for the pitching torque, we use the exact centroid of the wetted segment, which moves with the waterline through each contact and gives a more accurate torque (§3).

1 Setup

We assume the stone is a thin cylindrical disk with radius R, mass M, half-thickness e, spin $\dot{\phi}_0$ about the face normal, and that it is travelling along the (x, y) plane. The state of the stone is the planar position and velocity, *attitude* θ (tilt of the face above horizontal), and spin. Entry at incidence β below horizontal gives launch velocity $(V\cos\beta, -V\sin\beta)$.

The bounce occurs during a brief interval when the tilted lower edge sits below the local water height and the face is partially immersed over area S_{im}. Parameterising the wetted segment by the chord s along the face, with $u = 1 - s/R \in (-1, 1)$ the normalised waterline, this is the area of the circular segment cut from the disk by the waterline,

$$S_{\mathrm{im}}(s) = R^2\left[\arccos u - u\sqrt{1-u^2}\,\right], \qquad 0 \le s \le 2R, \tag{1}$$

which grows from zero at first impact, the full disk area once submerged, and dropped from the model when $s \ge 2R$. A wetted-edge depth y_{edge} maps to $s = y_{\mathrm{edge}}/\sin\theta$ (valid for θ not close to zero).

2 Hydrodynamic reaction

The wetted face experiences lift (normal to the face) and friction (along it), both quadratic in speed and linear in wetted area per [Bocquet 2003],

$$F = \tfrac{1}{2}\, C\, \rho_w\, V^2\, S_{\mathrm{im}}(s), \tag{2}$$

with water density ρ_w, speed $V = \|\mathbf{v}\|$, and coefficients $C_l, C_f = \mathcal{O}(1)$. Collecting $q = \frac{1}{2}\rho_w V^2 S_{\mathrm{im}}$ and resolving onto the axes, the friction opposing the in-plane motion is

$$M\,\ddot{x} = -\,q\,(C_l\sin\theta + C_f\cos\theta), \tag{3}$$

$$(M + m_a)\,\ddot{y} = -Mg + q\,(C_l\cos\theta - C_f\sin\theta) - \dot{y}\,\dot{m}_a. \tag{4}$$

A bounce occurs when the right side of (4) is net upward over the contact. (3) depletes V_x, and since lift $\propto V^2$, that loss eventually ends the run.

Added (virtual) mass. The lift from (2) alone is too weak to rebound the stone. The main force is inertial: as the stone pushes down, it accelerates a body of nearby water. The water resists being displaced, and its reaction back on the stone behaves like extra inertia carried along with it, which is equivalent to treating the stone as if it were heavier. We model this as an added mass proportional to the volume of fluid displaced [von Kármán 1930],

$$m_a = C_M\,\rho_w\,S_{\mathrm{im}}\,(2e), \tag{5}$$

which enters (4) twice: as augmented inertia $(M + m_a)$, and through the changing entrained momentum $m_a\dot{y}$ producing the reaction $-\dot{y}\,\dot{m}_a$ (the entry kick). By the chain rule through s,

$$\dot{m}_a = C_M\,\rho_w\,(2e)\,\frac{\mathrm{d}S_{\mathrm{im}}}{\mathrm{d}s}\,\dot{s}, \qquad \frac{\mathrm{d}S_{\mathrm{im}}}{\mathrm{d}s} = 2R\sqrt{1-u^2}, \quad \dot{s} = -\dot{y}/\sin\theta. \tag{6}$$

3 Evolving attitude

Since the stone is tilted when it hits the water, the force applied is uneven. If a shallow sliver is immersed, the energy is concentrated near the edge. Let P be the centroid of the immersed segment, ℓ is the distance from the centre O. The offset torques the disk toward steeper tilt (the wrong way) since the net vertical coefficient $C_l\cos\theta - C_f\sin\theta$ falls as θ grows, weakening the lift. This lever arm from O to the centre of pressure is

$$\ell = \frac{c^3}{12\,S_{\mathrm{im}}}, \qquad c = 2R\sqrt{1-u^2} \;\;\text{(wetted chord)}, \tag{7}$$

This is our one distinction from Bocquet, they fix $OP \sim R$, resolving ℓ with the waterline lets the torque track the contact rather than sit at a constant.

Either way it is modelled, spin stabilizes this through the Gyroscopic effect. Reducing Euler's equations for a spun disk [Bocquet 2003, Eq. 20] makes θ an oscillator with spin-set stiffness ω^2,

$$\ddot{\theta} + \omega^2(\theta - \theta_0) = \frac{M_\theta}{J_1}, \qquad \omega = \frac{J_0 - J_1}{J_1}\,\dot{\phi}_0, \tag{8}$$

with launch tilt θ_0, diametral and polar moments $J_1 = \frac{1}{4}MR^2$, $J_0 = 2J_1$ (so $\omega = \dot{\phi}_0$ for a thin disk), and pitching torque $M_\theta = \ell\,\langle F_y\rangle$ from the off-centre load, scaled by Bocquet's contact-average $\langle F_y\rangle \sim Mg$. The spin bounds the excursion to $\delta\theta \sim M_\theta/(J_1\omega^2)$, which diverges as $\dot{\phi}_0 \to 0$ (the disk tumbles). Bounding it recovers Bocquet's stability condition [Bocquet 2003, Eq. 21],

$$\dot{\phi}_0 \gg \sqrt{g/R}. \tag{9}$$

4 Coupled water surface

Since we don't need to model the subsurface flow, we model the water as a free surface using a height field $h(x, y, t)$. We model the surface ripples with a damped 2D wave equation PDE, and accelerate it on a GPU grid with an explicit leapfrog stencil,

$$\partial_t^2 h = c^2\nabla^2 h - \gamma\,\partial_t h, \qquad h_{ij}^{n+1} = \left(2h_{ij}^n - h_{ij}^{n-1} + C^2\,\nabla_h^2 h_{ij}^n\right)\kappa, \tag{10}$$

with wave speed c, damping γ, the discrete Laplacian ∇_h^2 a five-point stencil over grid spacing Δx, Courant number $C = c\,\Delta t/\Delta x < 1/\sqrt{2}$ for stability, and per-step damping factor $\kappa = 1 - \gamma\,\Delta t$.

Two-way coupling. Each contact deposits the exchanged vertical momentum $\Delta p = M(\dot{y}_{\text{out}} - \dot{y}_{\text{in}})$ as a Gaussian height impulse at the impact point (stone $\rightarrow$ water), and the disturbed height under the stone is read back as the level the next contact sees (water $\rightarrow$ stone), so waves generated will affect the skip (including possibly from a stone's own previous bounces).

5 Bounce-count bound

A stone survives only while it carries enough horizontal speed to generate lift, so the run ends once friction has drained V_x over a succession of bounces spaced ℓ_b apart. Counting those bounces gives Bocquet's closed-form upper bound on the skip count [Bocquet 2003, Eq. 23],

$$N_c = \frac{V_{x0}^2}{2\,g\,\mu\,\ell_b}, \qquad \ell_b = \frac{2\pi V_{x0}}{\omega_0}, \qquad \omega_0^2 = \frac{C\,\rho_w\,V_{x0}^2\,2R}{2\,M\sin\theta}, \tag{11}$$

with inter-bounce distance ℓ_b and contact angular frequency ω_0. Here $C = C_l\cos\theta - C_f\sin\theta$ is the net vertical coefficient, and $\mu = (C_l\sin\theta + C_f\cos\theta)/C$ is the friction-to-lift ratio. It notably treats each rebound as vertically elastic, it ignores the impact energy loss and *overestimates*. The simulator instead integrates (3)–(4) directly, which captures that loss; N_c is reported alongside the integrated outcome as an analytic upper bound for comparison.

References

[Bocquet 2003] Bocquet, L. 2003. The physics of stone skipping. *American Journal of Physics 71*, 2, 150–155.

[Bocquet et al. 2005] Bocquet, L., Rosellini, L., Hersen, F., and Clanet, C. 2005. Skipping stones. *Journal of Fluid Mechanics 543*, 137–146.

[Clanet et al. 2004] Clanet, C., Hersen, F., and Bocquet, L. 2004. Secrets of successful stone-skipping. *Nature 427*, 6969, 29.

[Nagahiro and Hayakawa 2004] Nagahiro, S., and Hayakawa, Y. 2004. A theoretical and numerical approach to the "magic angle" of stone skipping. *arXiv:physics/0411125*.

[von Kármán 1930] von Kármán, T. 1930. The impact on seaplane floats during landing. *NACA Technical Note 321*.